

Technology Stack

Date	1 July 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction System.
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit	Used to build an interactive and user-friendly web interface for entering country indicators and displaying HDI prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3	<i>Handles data preprocessing, machine learning prediction, and application logic using Scikit-learn.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	HDI.csv (CSV File)	<i>Stores the HDI dataset used for training and testing the prediction model. A CSV file is sufficient for this project.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Streamlit Community Cloud / Local Deployment	<i>Enables easy deployment of the application online or execution on a local machine during development.</i>
5	Version Control & CI/CD	Git & GitHub	<i>Used for source code management, version tracking, and collaboration among team members.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, Pickle</i>	<i>Scikit-learn is used for model training, Pandas and NumPy for data processing, and Pickle for saving/loading the trained model and label encoder.</i>